



ADVANCED MATHEMATICS

ACADEMIC

Grade 11/12
Curriculum Guide

ADVANCED MATHEMATICS

Prerequisite: Pre-Calculus

Course Description:

This course enhances learners' critical thinking, analytical reasoning, creativity, and problem-solving skills through construction, analysis, and interpretation of mathematical models drawn for real-life contexts. Learners investigate linear, quadratic, exponential, and trigonometric functions by representing them algebraically, graphically, and numerically to analyze patterns in data, distinguish among model types, and evaluate the appropriateness, reasonableness, and limitations of solutions in business, science, engineering, technology, and everyday applications. The course deepens understanding of trigonometric concepts, including working with angles and circular quantities, analyzing periodic behavior and spatial relationships, interpreting graphs, transforming expressions, and justifying solutions in varied real-life problems related to modeling and decision-making.

To broaden representational fluency, the course provides the option to extend mathematical understanding through the use of the polar coordinate system to locate and convert points, sketch polar curves, and model phenomena that are not effectively represented in rectangular coordinates.

Using appropriate technology, learners use digital and information literacy skills to visualize, analyze, and communicate results, as they collaboratively build mathematical models, including those involving trigonometric functions, while strengthening communication, initiative, adaptability and leadership. By applying mathematical concepts to authentic real-life scenarios, learners cultivate creativity, innovation, and responsible decision-making, preparing them for higher education, research, entrepreneurship, and active participation in a data-driven global society.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
NUMBER AND ALGEBRA (NA)	1. Basic Mathematical Modeling	1. differentiate between linear, quadratic, and exponential models based on data patterns or graphs. 2. interpret linear models in real-life contexts such as cost analysis, revenue prediction, or population growth. 3. interpret quadratic models in real-life contexts involving area, projectile motion, or optimization. 4. interpret exponential models in real-life contexts such as compound interest, population growth, or decay. 5. evaluate the appropriateness, reasonableness, and limitations of using linear, quadratic, or exponential models in a given real-life context.
	2. Angles and Their Measures	6. identify an angle by describing its initial and terminal sides and its direction of rotation (clockwise or counterclockwise). 7. sketch angles in standard position. 8. convert radian measures to degree measures and vice versa. 9. use radian measure to find the arc length and the area of a sector. 10. solve problems involving radian measure, arc length, and area of a sector.
	3. Trigonometric Functions	11. illustrate the different trigonometric functions in relation to the unit circle. 12. use reference angles to find values of trigonometric functions. 13. determine the domain and range of the different trigonometric functions. 14. graph the six trigonometric functions using features such as amplitude, period, phase shift, and asymptotes. 15. express trigonometric functions of acute angles as functions of the complement of the angle, using cofunction identities. 16. simplify trigonometric expressions by using Pythagorean identities.

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
		17. solve problems involving trigonometric functions including periodic phenomena.
Performance Standards The learners can analyze and model real-life situations using linear, quadratic, and exponential functions. They construct, evaluate, and verify these algebraically and graphically, apply function transformations using appropriate technology to represent and interpret solutions, and explain key function behaviors and limitations. The learners can also distinguish among these function types using data, graphs, and context, and evaluate the suitability and reasonableness of models in business, science, and everyday applications. They communicate conclusions using graphs, tables, equations, and written explanations to support sound decision-making. The learners can also model real-life phenomena using angle and radian measures, arc length, area of a sector, and trigonometric functions. They can present scenario analyses and clearly communicate their reasoning through visual and algebraic representations demonstrating how these concepts support problem-solving and informed decision-making in science, technology, and engineering.		
NUMBER AND ALGEBRA (NA)	4. Trigonometric Identities	18. apply sum and difference formulas for trigonometric functions to simplify expressions. 19. apply double-angle and half-angle formulas to find exact values or simplify trigonometric expressions. 20. prove trigonometric identities by applying the appropriate formulas. 21. solve problems involving trigonometric identities.
	5. Inverse Trigonometric Functions	22. illustrate the domain and range of the inverse trigonometric functions. 23. graph inverse trigonometric functions. 24. evaluate inverse trigonometric expressions. 25. solve problems involving inverse trigonometric functions.

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
	6. Polar Coordinates*	26.illustrate the polar coordinate system. 27.locate points in the polar coordinate system. 28.convert coordinates from rectangular to polar systems and vice versa. 29.identify polar curves from their equations. 30.sketch the graphs of polar equations. 31.solve problems involving the polar coordinate system.
Performance Standards The learners can use their understanding of trigonometric concepts to analyze, represent, and solve mathematical and real-life situations. They can apply key trigonometric formulas to simplify expressions, justify steps, and construct proofs, and evaluate and interpret trigonometric and inverse trigonometric functions using appropriate strategies. The learners communicate ideas through graphs, equations, tables, and explanations, showing ease in shifting between different representations. When solving contextual problems, they can apply inverse trigonometric functions to generate reasonable solutions and explain their significance. *To extend their mathematical understanding, the learners can use the polar coordinate system to describe and analyze relationships that are not effectively represented in rectangular coordinates. They can locate and convert points, identify and sketch polar curves, and use these tools to model real-world situations accurately. The learners can integrate their understanding of trigonometric identities, inverse functions, and coordinate systems to build meaningful models, explain reasoning effectively, and make informed decisions, demonstrating how abstract mathematical ideas connect to practical applications in science, technology, engineering, and everyday problem-solving.		

* This is an optional topic intended to enhance mathematical understanding and may be covered based on the learners' context.